

\$derived Rune

Q37

SVELTE 5 RUNES REACTIVITY OPTIMIZATION

ANSWER. The `$derived` rune calculates derived state synchronously from other state variables. When any of the source `$state` signals it depends on change, the `$derived` value automatically re-evaluates.

Its primary optimization benefit is memoization. If the underlying state hasn't changed, Svelte returns the cached value instead of re-running the computation. This prevents expensive recalculations and unnecessary DOM updates. Additionally, `$derived` tracks its dependencies dynamically at runtime, avoiding stale closures.

App.svelte

```
<script>
  let numbers = $state([1, 2, 3, 4, 5]);

  // Automatically recalculates only when 'numbers' changes
  let total = $derived(numbers.reduce((a, b) => a + b, 0));
</script>

<div>The total is {total}</div>
```

❶ **Summary.** Use `$derived` for computing values derived from `$state`. It offers automatic dependency tracking and memoization, recalculating only when necessary to optimize performance.

Input 1

`$state(A)`

Input 2

`$state(B)`

Computation
\$derived(A + B)
Memoized Evaluation
Output
Cached Result